Python Notes – Errors, Dynamic Typing & REPL

What is an Exception?

While writing programs, you may sometimes make mistakes.
Python detects many of these mistakes while the program is running and displays an exception (an error).
The exception name helps us understand what went wrong so that we can correct the program.



QUICK REFERENCE

Error	Meaning	Common cause	First thing to try
NameError	Name doesn't exist	Typo, variable not defined yet	Check spelling and define the variable before using it.
TypeError	Wrong type for an operation	Mixing str and int, wrong argument type	Use type() to check the data type. Convert using str(), int() or float() if needed.
AttributeError	The object doesn't have that method	Calling .upper() on an int, .append() on a str, etc.	Use dir() to see the available methods.

Tip: When Python displays an exception, read the last line first.
It usually tells you exactly what went wrong.

1. NameError

What it means:
Python doesn't know the name / variable that you used.

✗ Incorrect example

```
print(total)
```
(total not defined)

Output

```
NameError: name 'total' is not defined
```

✓ Correct example

```
total = 450
print(total)
```

Output

```
450
```

Key point:
Use a variable only after you have defined it and spelled it correctly.

2. TypeError

What it means:
The operation or function is used with a wrong data type.

✗ Incorrect example

```
age = 12
print("Age: " + age)
```
(str + int)

Output

```
TypeError: can only concatenate
str (not "int") to str
```

✓ Correct example

```
age = 12
print("Age: " + str(age))
```

Output

```
Age: 12
```

Key point:
Make sure the data types are correct for the operation. Convert if needed.

3. AttributeError

What it means:
The object (of a particular type) does not have the method or attribute you called.

✗ Incorrect example

```
x = 10
x.upper()      (int has no upper())
```

Output

```
AttributeError: 'int' object has no
attribute 'upper'
```

✓ Correct example

```
s = "hello"
print(s.upper())
```

Output

```
HELLO
```

Key point:
Different types have different methods.
Use methods that are available for that type.

USEFUL PYTHON FUNCTIONS

Function	What it does
type(x)	Returns the type of x.
dir(x)	Lists the methods and attributes available for x.
str(x)	Converts x to a string.
int(x)	Converts x to an integer.
float(x)	Converts x to a float.

You can use these functions to understand the data you are working with and fix errors easily!

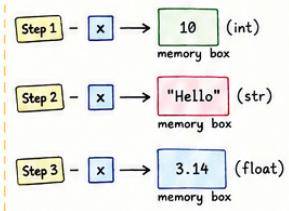
4. Dynamic Typing

A variable is simply a name that refers to a value.
The value has a type, not the variable.

Example

```
x = 10
print(type(x))      # int
x = "Hello"
print(type(x))      # str
x = 3.14
print(type(x))      # float
```

How it works (inside Python)



Remember
- A variable can store different types of values at different times.
- Python automatically keeps track of the type of the current value.
- This feature is called Dynamic Typing.

5. REPL

REPL stands for Read, Evaluate, Print, Loop.

The Python Shell works like a REPL.
It reads your command, executes it, prints the result, and waits for the next command.
It is useful for quick experiments and testing small pieces of code.

Example (Python Shell)

```
>>> 2 + 3
5
>>> "hi".upper()
'HI'
>>>
```

Type a command, see the result, try something else!

Created with the help of ChatGPT by Mandar Vaze